Lucid: Six-Tier Decoupled System Architecture

Modular Enterprise Architecture for Goal-Based Retail Wealth Management

TIER 1: CLIENT PRESENTATION LAYER (Next.js 16.4 • React 19 • Tailwind CSS)

PORT 3000

- **Investor Onboarding** — 5-Question Risk & Mandate
- **Portfolio Dashboard** — Whole Shares & NAV Charts
- **2-Level Stock-Why Modal** — Plain-English & Exchange Links
- **Demat CAS & Drift Radar** — HHI Audit & Trade Previews

REST APIs / JWT Session / Strict CORS

TIER 2: BACKEND API GATEWAY & SERVICES (FastAPI 0.115 • Uvicorn ASGI)

82 REST ENDPOINTS

- **Auth & Rate Limiter** — Argon2id Hash • 120 req/min
- **Portfolio & Why Services** — Creation, Versioning & Proofs
- **Demat Diagnosis Service** — pdfplumber CAS Parser & HHI
- **300s Drift Monitoring** — Session Lock 847291 • Alerts

TIER 3: AGENTIC AI INTELLIGENCE LAYER

LangGraph 0.4

PortfolioAgentState (35 Fields across 5 Partitions)
Mandate • Universe • Analyst Reports • Critic Loop • Math

- **Macro (N3)** — Repo & CPI — Cash Buffer
- **Market (N4)** — VIX & 200 DMA — Beta Ceiling
- **Fundam. (N5)** — ROE, P/E, D/E — Z-Factor Score
- **News/SEBI (N6)** — 30d Headlines — Pledges <5%

- **Lead Strategist (Node 7)** — Selects K ∈ [10..16] Candidate Stocks
- **Adversarial Critic (N8-N9)** — Rework Loop (≤2) • Mandate Audit

Primary: Gemini 3.8 Flash ⇄ 0.48s Fallback: Rule Engine v2

10-16

TIER 4: DETERMINISTIC MATH & FINANCIAL ENGINE

Zero-LLM • SciPy 1.15

SciPy SLSQP Constrained Quadratic Optimizer
Max Sharpe (Mod/Agg) | Min Var (Cons) • Bounds: Stock ≤25%, Sector ≤40%

Whole-Share Integer Lot Sizing & Cash Conservation
Floor: Qi = ⌊(wi* · V) / Pi⌋ • Greedy Remainder Pass • Guarantee: Cash ≥ ₹0

Statutory Indian Transaction Cost Engine (Oct 2024 SEBI)
STT (0.10%) • Exchange (0.00297%) • Stamp (0.015%) • GST (18%)

Invariant Gate (N11): Post-Rounding Validation & Metric Verification

TIER 5: RELATIONAL PERSISTENCE LAYER (PostgreSQL 16.4 • SQLAlchemy 2.0 • Alembic)

62 DOMAIN TABLES • ACID COMMIT

- **Security Master (9 Tables)** — 2,639 Stocks • 2,715 Listings
- **Market & News (11 Tables)** — Prices, OHLCV, Z-Scores, VIX
- **Portfolios & Why (13 Tables)** — Versions, Decisions, Evidence
- **Audit & Demat (17 Tables)** — Agent Runs, CAS Audits, Alerts

Ingestion Adapters • Resilient Caching

TIER 6: EXTERNAL MARKET DATA & CLOUD INTEGRATIONS

FREE-TIER ADAPTERS & RESILIENT CACHE

- **NSE & BSE Archives** — 2,639 Master Equities
- **Yahoo Finance API** — 15-Min Lag • 1Y OHLCV
- **RBI Bulletins & MoSPI** — Repo 6.50% • CPI 5.10%
- **NSE Index Stream** — India VIX & Nifty 50
- **Google Gemini API** — gemini-3.8-flash